```
#include <Servo.h>
#include <NewPing.h>

#define TRIGGER_PIN 12
#define ECHO_PIN 11
#define MAX_DISTANCE 300

#define GROUND_JOY_PIN A3 // joystick ground pin will connect to Arduino analog pin A3
#define VOUT_JOY_PIN A2 // joystick +5 V pin will connect to Arduino analog pin A2
#define XJOY_PIN A1 // X axis reading from joystick will go into analog pin A1

#define CloseDistance 6 //the distance the servo motor is from the ground when the base of the claw is touching the ground
#define closedAngle 110 //angle the servo motor is at when the claw is CLOSED
#define openAngle 75 //angle the servo motor is at when the claw is OPEN
#define TimeRemainingClosed 4000UL //initial time remaining closed after the claw closes (until it gets to a certain height)
#define HeightStaysClosed 20 //cm which when the claw is above this height above the ground, it will stay closed no matter what

#define JoystickRestPos 516 //this varies with each joystick and should be tested before hand -- current one is from Mel & Emma's arduino kit
#define JoystickRange 50 //how far away the value of the joystick needs to get in order for the emergency open to be triggered

Servo myservo;

NewPing sonar (TRIGGER_PIN, ECHO_PIN, MAX_DISTANCE);

int pos = 0;    // variable to store the servo position

unsigned long TimeOfClose = 0;

bool closed = false;

int target = openAngle;
```

```
void setup() {
  myservo.attach(8);  // attaches the servo on pin 8 to the Servo object

  pinMode (10, OUTPUT);
  pinMode (13, OUTPUT);

  digitalWrite(10,LOW);
  digitalWrite(13, HIGH);

  Serial.begin(9600);

  myservo.write(openAngle);

  pinMode(VOUT_JOY_PIN, OUTPUT); //pin A3 shall be used as output
  pinMode(GROUND_JOY_PIN, OUTPUT) ; //pin A2 shall be used as output
  digitalWrite(VOUT_JOY_PIN, HIGH) ; //set pin A3 to high (+5)
  digitalWrite(GROUND_JOY_PIN,LOW) ; //set pin Ad to low (ground)
}

void loop() {
  unsigned long now = millis();
  int joystickXVal = analogRead(XJOY_PIN) ; //read joystick input on pin
A1. Will return a value between 0 and 1023.

  int DISTANCE_IN_CM = sonar.ping_cm(); //reads the sonar sensor

  bool valid = (DISTANCE_IN_CM > 0); //ensures that the sonar doesn't
misinterpret no reading as being very close

  target = closed ? closedAngle : openAngle; //if closed is true, keep
target at closedAngle, otherwise, keep it at openAngle

  Serial.print("Ping: ");
  Serial.print(DISTANCE_IN_CM);
  Serial.println("cm");

  if (valid && DISTANCE_IN_CM <= CloseDistance && closed == false){
      target = closedAngle;
```

```
      closed = true;

      TimeOfClose = now;
    }

  if (closed == true && (now - TimeOfClose) < TimeRemainingClosed){
    target = closedAngle;
  }

  if (closed == true && (now - TimeOfClose) >= TimeRemainingClosed){
    if (valid && DISTANCE_IN_CM > HeightStaysClosed){
      target = closedAngle;

      closed = true;
    }

    if (valid && DISTANCE_IN_CM <= HeightStaysClosed){
      target = openAngle;

      closed = false;
    }
  }

  //any movement on the joystick will cause the servo motor to immediately
open:
  if (joystickXVal < (JoystickRestPos - JoystickRange) || joystickXVal >
(JoystickRestPos + JoystickRange)){
    target = openAngle; //immediately opens the claw no matter where in
the pickup cycle it is

    closed = false;
  }

  myservo.write(target);

}
```